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Removing Cost Volumes from Optical Flow Estimators

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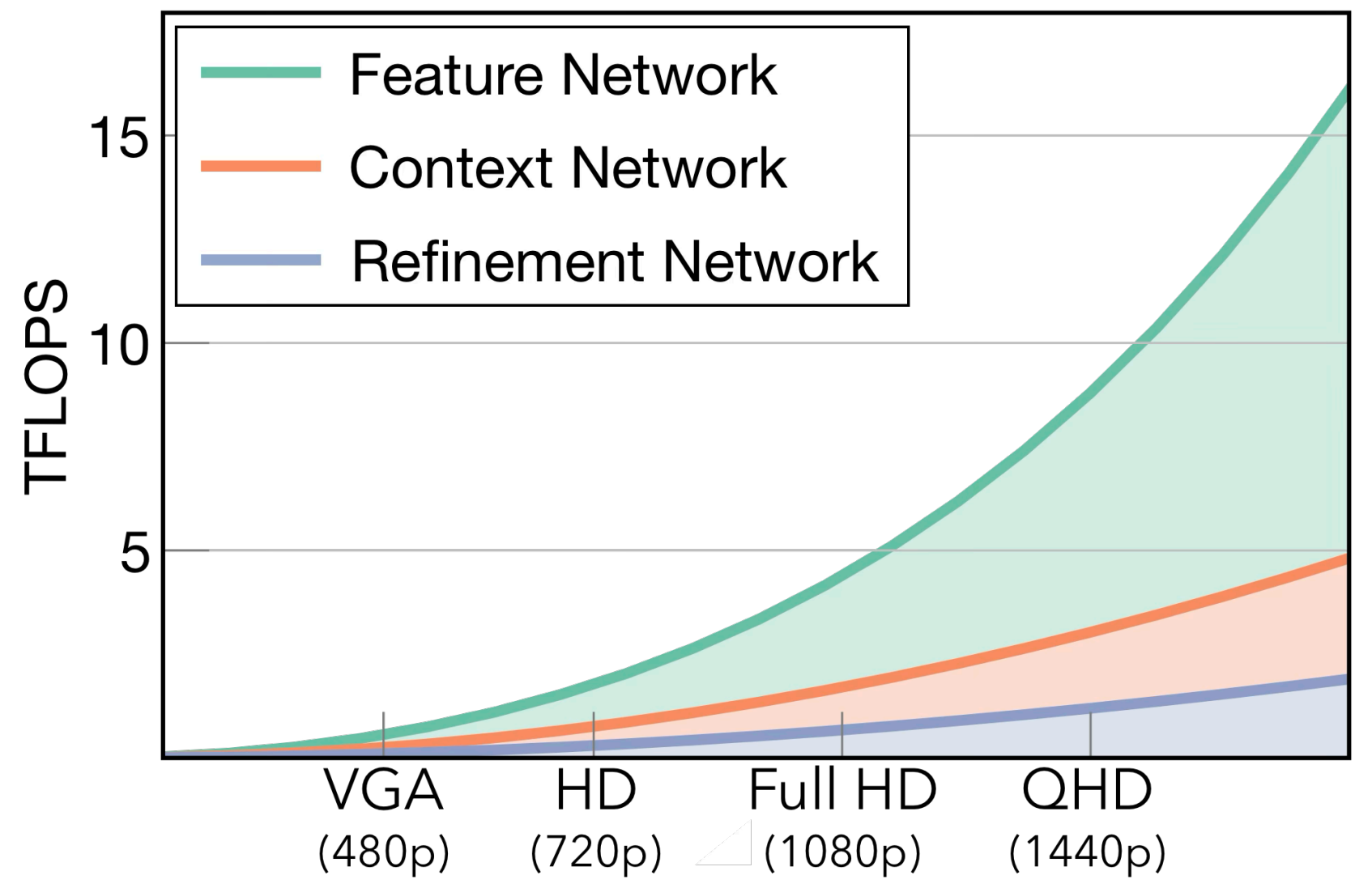
Project Page

Motivation

- All modern optical flow architectures use some form of cost volume (e.g., as part of the feature network).
- Cost volumes are expensive to compute and are, therefore, often a limiting factor regarding speed and input resolutions.

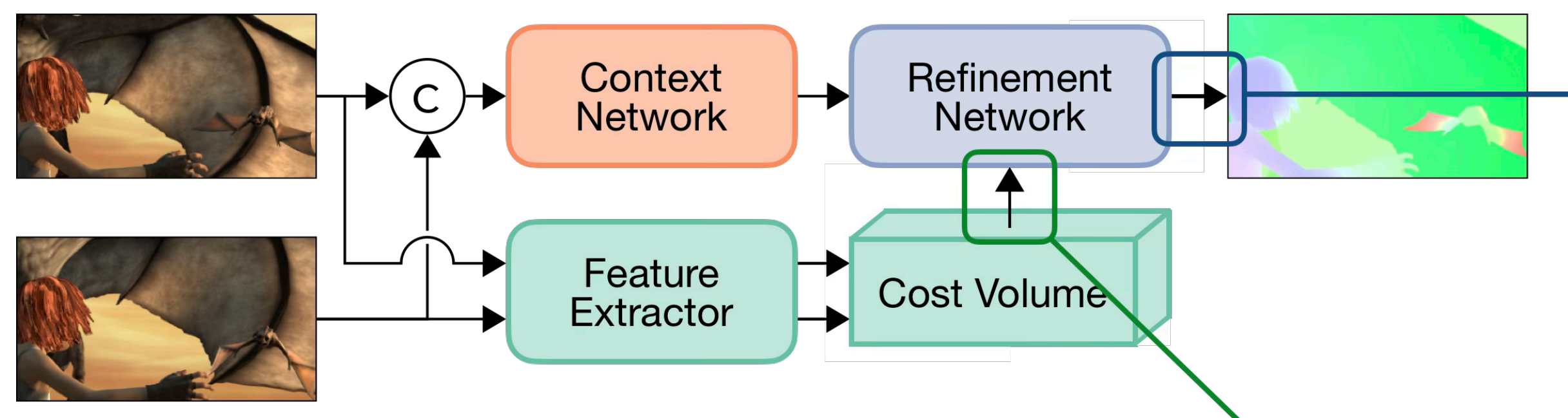


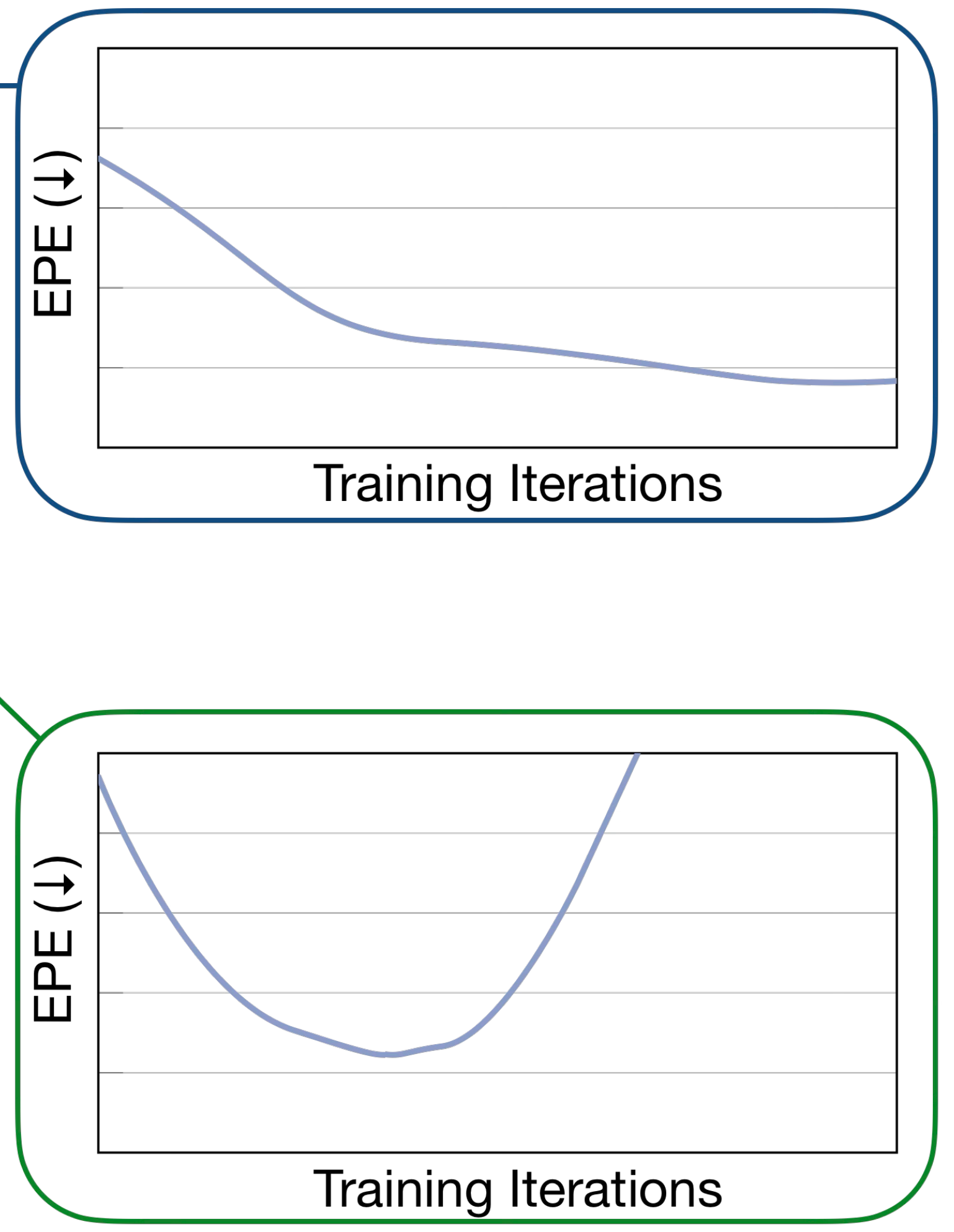
Contributions

We present **ReCoVEr** – A training strategy for removing the cost volume from an optical flow estimator during training.

- We find that cost volumes are only required during the early phase of the training.
- As a result, we achieve a significant reduction in memory and compute requirements while reaching SOTA accuracies.

Tale of a Failed Experiment



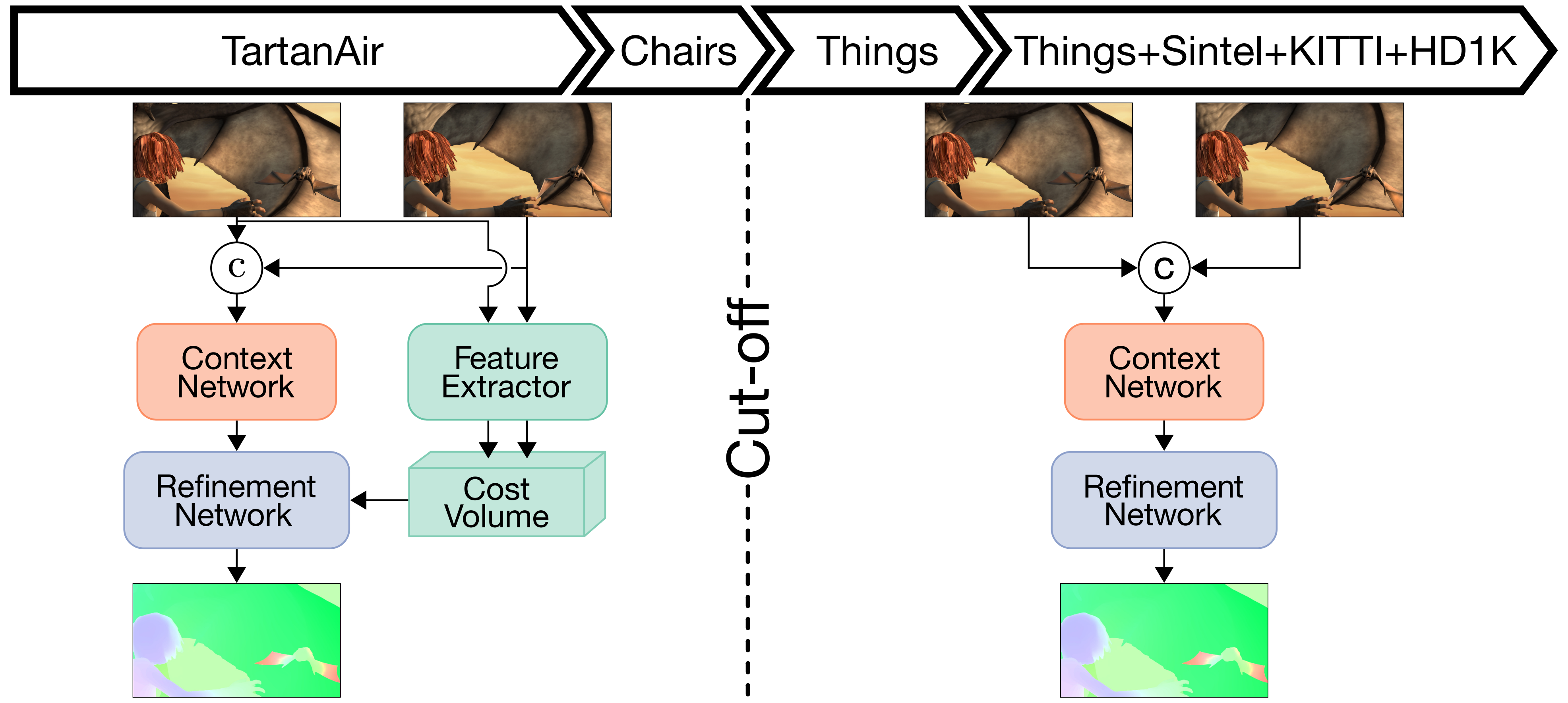


We initially experimented with more efficient feature extractors and obtained "weird" results:

- Reduced EPE during training.
- But: Increased EPE from cost volume.

→ Low end-point error seems to be possible without cost volumes.

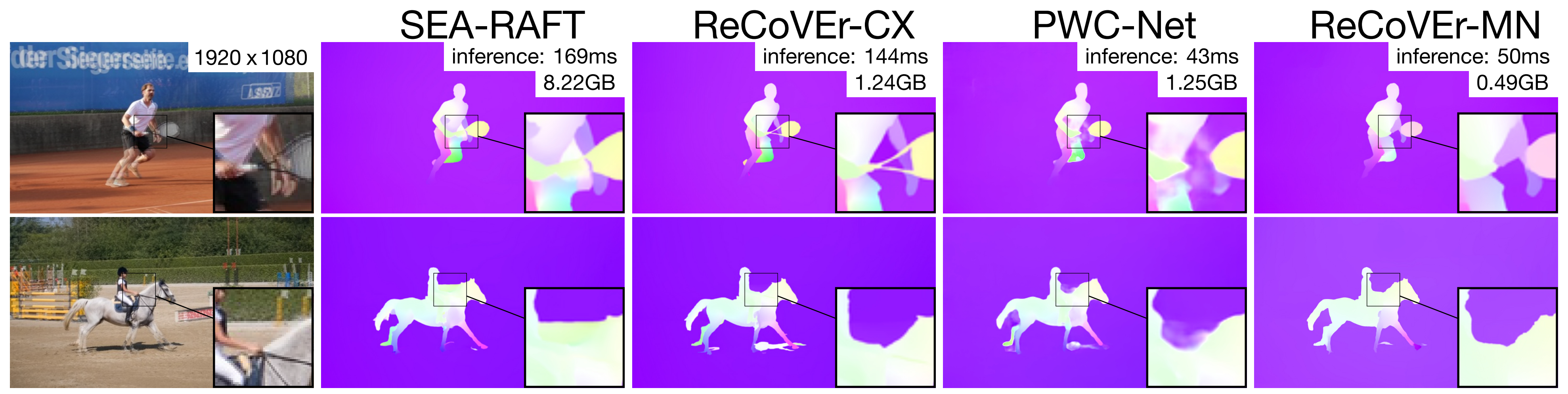
Training Schedule



Step-by-step:

1. Start training with a cost volume.
2. Remove cost volume after a certain number of training steps.
3. Continue training the resulting architecture.

Comparisons



References & Acknowledgments

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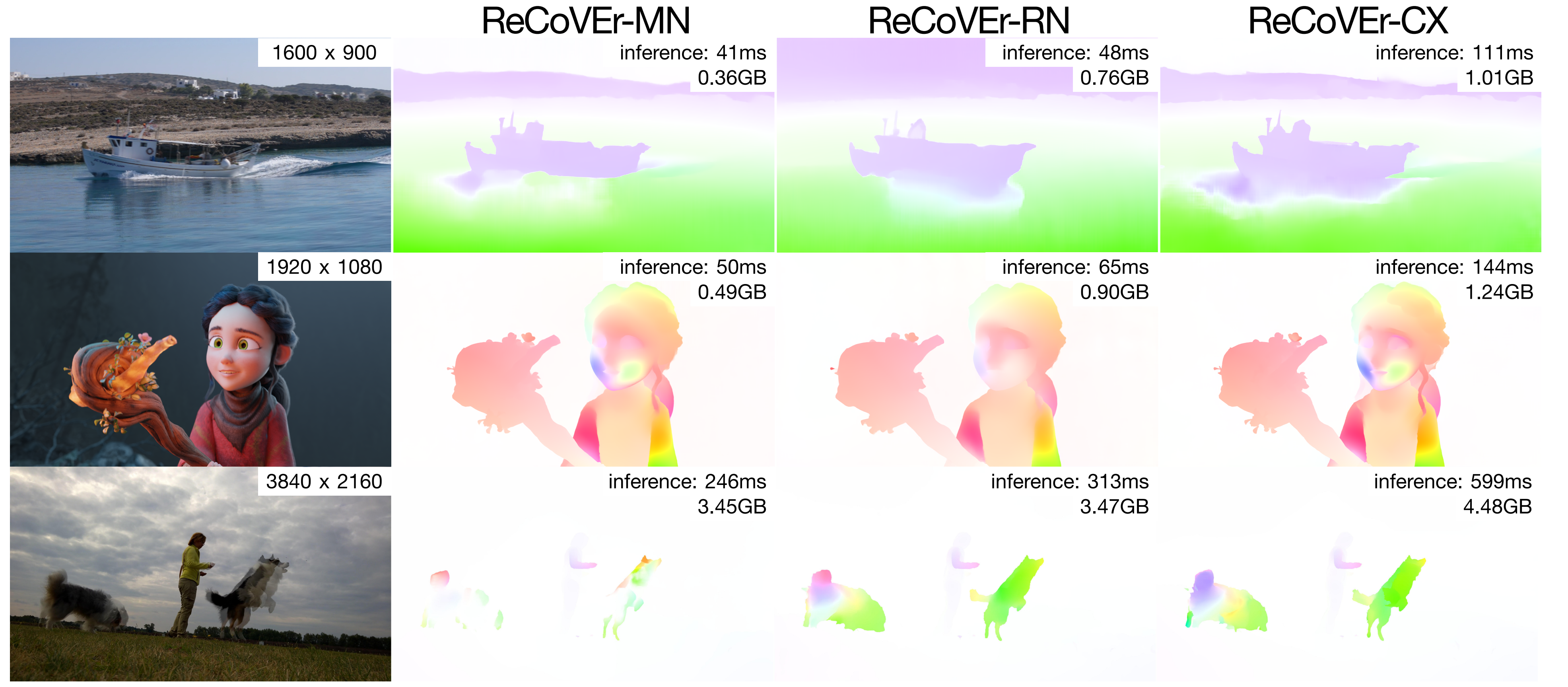
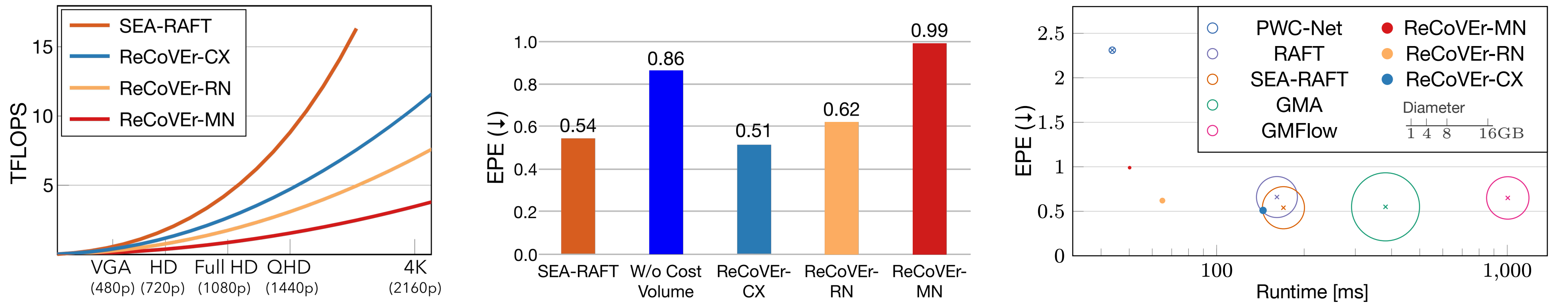
erc European Research Council Established by the European Commission

Qualitative Results



Evaluation on Spring Dataset

Our simple training procedure leads to



Different trade-offs between accuracy and compute requirements

State-of-the-art accuracy with reduced memory footprints and runtimes

Method	Sintel	TartanAir	total	low-detail	high-detail	matched	unmatched	rigid	non-rigid	s0-10	s10-40	s40+	time[ms]	memory[GB]
PWC-Net	✗	✗	2.31	2.27	10.84	2.13	11.28	2.10	4.45	1.83	3.16	15.87	**43.72**	1.25
RAFT	✓	✗	0.66	0.61	9.66	0.54	6.55	0.24	4.91	0.18	2.85	10.83	161.19	8.00
GMFlow	✓	✗	0.65	0.61	8.93	0.54	5.99	0.41	3.04	0.36	1.36	**8.28**	1004.23	8.28
GMA	✓	✗	0.55	0.50	11.35	0.45	**5.97**	**0.20**	4.13	**0.16**	1.82	10.33	381.21	13.29
SEA-RAFT	✓	✓	0.54	0.50	**8.76**	0.40	7.25	0.31	**2.90**	0.20	**0.95**	10.56	169.72	8.22
ReCoVEr-MN	✓	✓	0.99	0.93	14.36	0.82	9.48	0.54	5.57	0.28	1.95	21.96	50.16	**0.49**
ReCoVEr-RN	✓	✓	0.62	0.59	9.48	0.49	7.77	0.34	3.60	0.18	1.24	13.78	65.03	0.93
ReCoVEr-CX	✓	✓	**0.51**	**0.47**	9.10	**0.39**	6.51	0.24	3.28	**0.16**	1.08	10.80	144.41	1.24